PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T47 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 22 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T47 • Expertise Q16+ • 5 items • 22 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Area & Volume of Similar Shapes

Unit 2 • Expertise • 22 marks • 5 item(s)

June 2025 | 4MA1-2H Q26 | 4 marks

Solve a volume difference or sum for similar solids

November 2023 | 4MA1-1H Q23 | 5 marks

Convert between area and volume ratios

November 2023 | 4MA1-2H Q23 | 5 marks

Use similar solids to solve a frustum problem

January 2022 | 4MA1-1H Q17 | 4 marks

Solve a volume difference or sum for similar solids

January 2023 | 2H Q19 | 4 marks

Solve an area difference or sum for similar figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** $=$ 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

WORKING SPACE

Volumes of similar solids **4 marks**

Scale volumes

$$A : B = 31 : 18.6 = 5 : 3 \Rightarrow V_A : V_B = 5^3 : 3^3 = 125 : 27.$$

Use the difference

$$V_A = 125k,\ V_B = 27k; \quad 98k = 735 \Rightarrow k = 7.5.$$

Find A

$$V_A = 125(7.5) = 937.5\ \mathrm{cm}^3.$$

Final answer

$$937.5\ \mathrm{cm}^3\ (\text{or } 938\ \mathrm{cm}^3)$$

4MA1-1H Q23 demand **5 marks**

23 Solid **A** is similar to solid **B**

Here is some information about solid **A** and solid **B**

	solid **A**	solid **B**
Height (cm)	3^x	
Area (cm^2)	7776	486
Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid **B**
Give your answer as a decimal.

.... cm

WORKING SPACE

4MA1-1H Q23 demand **5 marks**

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	solid **A**	solid **B**
Height (cm)	3^x	
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Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid **B**
Give your answer as a decimal.

.... cm

Similar solids: height of B **5 marks**

Demand – Similar solids: height of B

Find the linear scale factor

$$\frac{A_A}{A_B} = \frac{7776}{486} = 16 \Rightarrow k = 4.$$

Use the volume ratio

$$\frac{8^x}{2^{x+4}} = k^3 = 64 \Rightarrow 2^{2x-4} = 2^6 \Rightarrow x = 5.$$

Scale the height

$$h_A = 3^5 = 243, \quad h_B = 243/4 = 60.75 \text{ cm}.$$

Final answer

60.75 cm

4MA1-2H Q23 whole **5 marks**

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

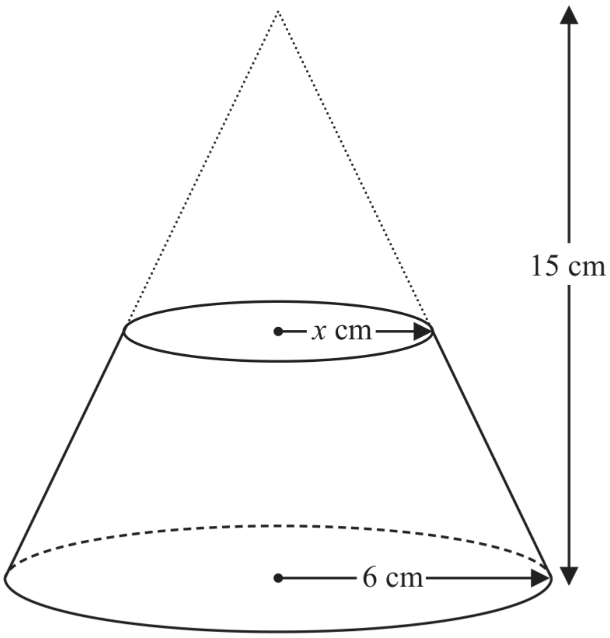

Diagram **NOT** accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.
The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

$x =$ ....

WORKING SPACE

Continue your method **5 marks**

WORKING SPACE

4MA1-2H Q23 whole **5 marks**

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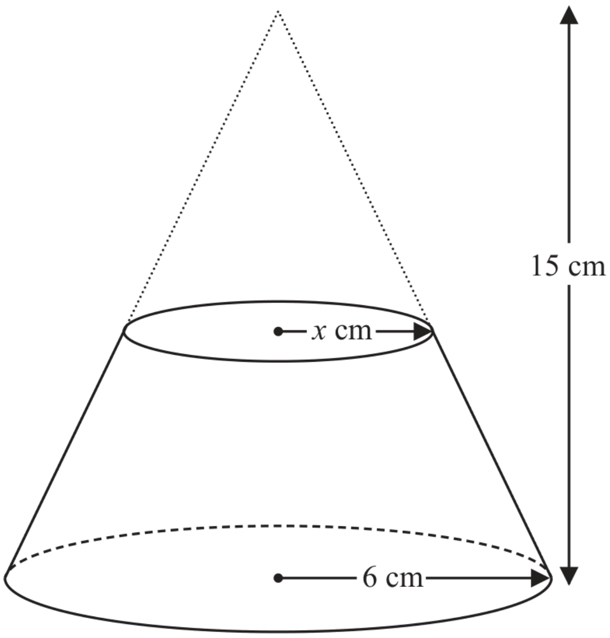

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Show clear algebraic working.

$x =$ ....

Radius of a similar-cone frustum **5 marks**

whole – Radius of a similar-cone frustum

Find the large-cone volume

$$V_L = \frac{1}{3}\pi(6^2)(15) = 180\pi.$$

Use similarity

$$\text{The small cone has height } \frac{x}{6}(15) = \frac{5}{2}x, \text{ so its volume is } \frac{1}{3}\pi x^2\left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3.$$

Use the frustum volume

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi \Rightarrow \frac{5}{6}x^3 = \frac{288}{25}.$$

Solve the cubic

$$x^3 = \frac{1728}{125} = \left(\frac{12}{5}\right)^3 \Rightarrow x = \frac{12}{5} = 2.4 \text{ cm}.$$

Final answer

x = 2.4 cm

4MA1-1H Q17 **4 marks**

17 **A** and **B** are two similar vases.

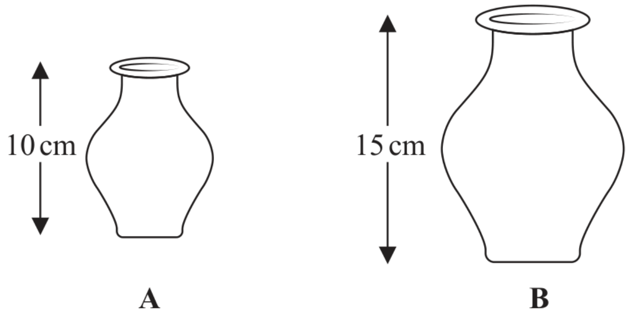


Diagram **NOT** accurately drawn

Vase **A** has height 10 cm.
Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

.... cm^3

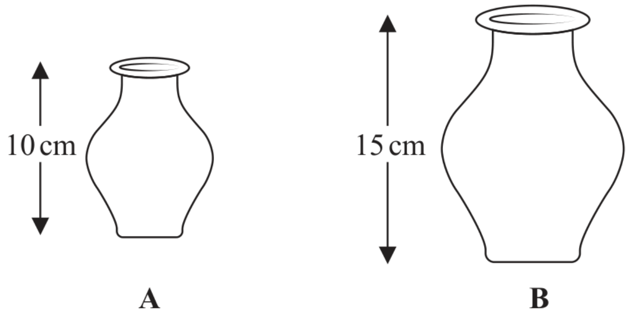
Continue your method **4 marks**

WORKING SPACE

4MA1-1H Q17

4 marks

17 **A** and **B** are two similar vases.



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Calculate the volume of vase **A**

.... cm^3

Chapter: Area & Volume of Similar Shapes **4 marks**

Q17 – Chapter: Area & Volume of Similar Shapes

Family: Use volume scale factors of similar solids • Variant: Solve a volume difference or sum for similar solids

Method
The linear scale factor from A to B is 15/10=1.5, so the volume scale factor is 1.5^3=27/8.
Why: Similar solids scale volumes by the cube of the linear factor.

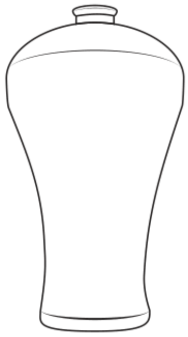
Method
If V is the volume of A, (27/8)V−V=1197.
Why: Use the stated volume difference.

Method
(19/8)V=1197, so V=1197×8/19.
Why: Solve the linear equation in V.

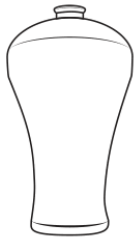
Method
V=504 cm^3.
Why: Evaluate and state the volume of vase A.

Final answer
volume of A: 504 cm^3

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT** accurately drawn

The vases are such that

$$\text{surface area of vase } \mathbf{B} = \frac{25}{64} \times \text{surface area of vase } \mathbf{A}$$

and that

$$\text{volume of vase } \mathbf{A} - \text{volume of vase } \mathbf{B} = 541.8\,\text{cm}^3$$

Calculate the volume of vase **B**

.... cm^3

(Total for Question 19 is 4 marks)

Answer reference | January 2023 | Question 19

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Question	Working	Answer	Mark	Notes
19	eg $\sqrt{\frac{25}{64}}\left(=\frac{5}{8}=0.625\right)$ **or** $\sqrt{\frac{64}{25}}\left(=\frac{8}{5}=1.6\right)$ **or** $\sqrt{25}:\sqrt{64}$ $(5:8)$ **or** $\sqrt{64}:\sqrt{25}$ $(8:5)$ **or** $\frac{(\sqrt{25})^3}{(\sqrt{64})^3}=\left(\frac{125}{512}=0.244140625\right)$ oe **or** $\frac{512}{125}=4.096$ $\frac{25^3}{64^3}=\frac{(\text{vol of }\mathbf{B})^2}{(\text{vol of }\mathbf{B}+541.8)^2}$ **or** $\frac{25}{64}=\frac{(\text{vol of }\mathbf{B})^{\frac{2}{3}}}{(\text{vol of }\mathbf{B}+541.8)^{\frac{2}{3}}}$ oe		4	M1 for a correct scale factor for length – may be given as a fraction or decimal or ratio **or** a correct scale factor for volume given as a fraction or decimal or ratio **or** a correct equation for the volume of vase **B**
	eg $\mathbf{B}\left(\frac{512}{125}-1\right)=541.8$ **or** $3.096\mathbf{B}=541.8$ oe **or** eg $\mathbf{A}\left(1-\frac{125}{512}\right)=541.8$ **or** $\frac{387}{512}\mathbf{A}=541.8$ $(8^3)k-(5^3)k(=387k)=541.8$ **or** eg $(k=)\frac{541.8}{387}\left(=\frac{7}{5}\right)$			M1 For a correct equation for the volume of **B** or a correct equation for the volume of **A**
	eg $(\mathbf{B})\,541.8\div"\frac{387}{125}"$ **or** 541.8 ÷"3.096" **or** eg $125\times"\frac{7}{5}"$ or $(\mathbf{A})\,541.8\div"\frac{387}{512}"(=716.8)$ oe			M1 For a completely correct method to find the volume of vase **B** or vase **A**
	Correct answer scores full marks (unless from obvious incorrect working)	175		A1 cao
				Total 4 marks

Worked solution

January 2023 | Question 19 | 4 marks

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Official final mark-scheme pages are embedded immediately after this question.